

Beware: conditions that masquerade as sports injuries

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Belgian decathlete Thomas Van der Plaetsen had testicular cancer diagnosed because abnormal levels of the HCG hormone were detected on a routine doping test in 2014. After treatment, 2016 saw him win the European Championships and compete at the Rio 2016 Olympic Games.

Not every patient who presents to the sport and exercise medicine clinician has a sports-related condition. Sport and exercise medicine has its share of conditions that must not be missed—‘red flag’ conditions that may appear at first to be rather benign. For example, a patient diagnosed with a minor ‘calf strain’ may have a deep venous thrombosis or a young basketball player with proximal anterior tibial pain who has been labelled as having Osgood-Schlatter disease may have an osteosarcoma or giant cell tumour of the proximal tibia. In this chapter we:

- outline a clinical approach that should maximise your chances of recognising a condition that is ‘masquerading’ as a sports-related condition
- describe some of these conditions to illustrate how they can present.

HOW TO RECOGNISE A CONDITION MASQUERADING AS A SPORTS INJURY

Three key factors will assist the clinician to avoid missing an atypical or masquerading condition. The first is keeping an open mind to the possibility. Too often, we become more rigid regarding our initial diagnosis rather than being open to consider other possibilities. Sometimes a fresh perspective from a colleague can be helpful in this regard.

The second and perhaps most important key is obtaining a detailed and complete history and physical examination. If you do not ask the question or consider a broader differential diagnosis, it is impossible to pick up the subtle clues pointing to a less likely but masquerading condition. Indeed, if the concept of a masquerading condition is not triggered from the history and examination, it is unlikely that appropriate investigations to make the diagnosis will be ordered. For example, if a patient presents with tibial

pain and it is, in fact, due to hypercalcaemia secondary to lung cancer, a bone scan of the tibia looking for stress fracture will usually not help with the diagnosis, but a history of weight loss, occasional haemoptysis and associated abdominal pain may. In a basketball player with shoulder pain, the history of associated arm tightness and the physical finding of prominent superficial veins are important clues to axillary vein thrombosis; a grey-scale ultrasound scan looking for rotator cuff tendinopathy will not provide the diagnosis.

PRACTICE PEARL

In sports medicine, it may seem simple to jump to the obvious conclusion regarding a specific diagnosis but if more in-depth questions or a thorough physical exam are not completed, it is difficult to be alert to a less common or ‘non-obvious’ diagnosis.

A third important key that points to masquerading conditions is when the clinical progression does not fit the pattern expected of the more common diagnosis or condition. When this occurs, consider alternative less common conditions. For example, the lingering contusion that should have resolved long ago could represent an underlying blood dyscrasia or tumour of the soft tissue or bone.

PRACTICE PEARL

You must ask yourself, ‘Could this be a rare condition or unusual manifestation?’ When the question is asked, then other options are entertained and the appropriate diagnosis can be conceived.

Ultimately, successful diagnosis of masquerading conditions requires keeping an open mind, performing a complete history and exam, and recognition of discrepancies between the patient's clinical features and clinical progress compared to what should be expected by the more typical diagnosis.

Conditions masquerading as sports injuries

Table 7.1 lists some of the conditions that may masquerade as sport and exercise medicine conditions. These are outlined below.

Bone and soft tissue tumours

Primary malignant tumours of bone and soft tissues are rare but when they occur it is most likely to be in the younger age group (second to third decade). Osteosarcoma can present at the distal or proximal end of long bones, more commonly in the lower limb, producing joint pain. Patients often recognise that pain is aggravated by activity and hence present to the sports medicine clinic. The pathological diagnosis of osteosarcoma is dependent on the detection of tumour-producing bone and so an X-ray may reveal a moth-eaten appearance with new bone formation in the soft tissues and lifting of the periosteum (Fig. 7.1). In young patients, the differential diagnosis includes osteomyelitis. It is recommended that any child or adolescent with bone pain be X-rayed.

Another presentation of tumour is when a patient presents with a mild injury that won't get better or with

an atypical finding on radiographs taken when a fracture occurs. If fractures occur with relatively low energy or when they do not have an appropriate mechanism of injury masqueraders should be considered. They may also present as progressive pain rather than pain that improves with rest and treatment.

Malignant tumours (e.g. of the breast, lung and prostate) may metastasise to bone. Patients may not recognise that a previously treated malignancy could be related to limb pain. Breast carcinoma may also present as a frozen shoulder. An accurate history is, therefore, central to making an accurate diagnosis. Red flag signs for malignancy or infection include prominent night pain, often being woken at night with pain, fever, loss of appetite, weight loss and malaise. Patients exhibiting these symptoms should be examined and investigated thoroughly to determine the cause. In the sports population, patients with metastatic disease are more likely older and have more complex medical histories.

Giant cell tumours, aneurismal bone cysts and unicameral bone cysts are benign or less aggressive bone tumours that can present in a similar fashion with pain and failure to resolve in a time frame expected of a simple contusion or traumatic injury. Osteoid osteoma is a benign bone tumour that often presents as exercise-related bone pain and tenderness and is frequently misdiagnosed as a stress fracture. It can occasionally be seen on plain radiographs (Fig. 7.2a). The bone scan appearance is similar to that of a stress fracture, although the isotope uptake is more intense and widespread. This condition is characterised clinically by the presence of night pain and by the abolition of symptoms with the use of aspirin. The tumour has a characteristic appearance on CT scan (Fig. 7.2b) with a central nidus.

Ultimately, like malignant tumours of bone, the examiner should be vigilant regarding atypical complaints and failure to progress along normal healing timelines expected of simple contusions, stress fractures or frank fractures. Care should be taken to thoroughly interpret all radiographs for other potential causes of bone weakening that led to a fracture in the first place.

As for atypical bone pathologies, when soft tissue lesions present with an unusual history or do not progress as expected, the clinician should readily consider a broader differential diagnosis.

Synovial cell sarcoma frequently involves the larger lower joints such as the knee and ankle. Patients present with pain, often at night or with activity, maybe with instability and swelling. As a masquerader, these present as chronic swelling and pain that does not resolve with standard treatment. Like synovial sarcoma, synovial chondromatosis and pigmented villonodular synovitis are benign tumours of the synovium¹ found mainly in



Figure 7.1 X-ray of an osteosarcoma in the distal femur

Table 7.1 Conditions that may masquerade as sport and exercise medicine conditions

CONDITION	
Bone and soft tissue tumours	• Osteosarcoma • Synovial sarcoma • Synovial chondromatosis • Pigmented villonodular synovitis • Rhabdomyosarcoma • Osteoid osteoma • Ganglion cyst
Rheumatological	• Inflammatory monoarthritis • Inflammatory polyarthritis • Inflammatory low back pain (e.g. sacroiliitis) • Entesopathies (e.g. psoriatic, reactive arthritis)
Disorders of muscle	• Dermatomyositis • Polymyositis • Muscular dystrophy
Endocrine	• Dysthyroidism • Hypercalcaemia • Hypocalcaemia • Hyperparathyroidism • Diabetes • Cushing's syndrome • Acromegaly
Vascular	• Venous thrombosis (e.g. deep venous thrombosis, axillary vein thrombosis) • Artery entrapment (e.g. popliteal artery entrapment) • Peripheral vascular disease
Genetic	• Marfan syndrome • Haemochromatosis
Granulomatous diseases	• Tuberculosis • Sarcoidosis
Infection	• Osteomyelitis • Septic arthritis • Shingles • Lyme disease
Regional pain syndromes	• Complex regional pain syndrome • Fibromyalgia/myofascial pain syndrome



(a)



(b)

Figure 7.2 Osteoid osteoma (a) Radiograph showing sclerotic focus in distal fourth metatarsal (b) Transverse image from a CT scan demonstrating an osteoid osteoma. Note the lytic lesion in the anterior cortex of the femoral neck (arrow), with radiofrequency probe in position for ablation

the knee, which present with mechanical symptoms, persistent swelling, and failure to respond to basic rest, ice and anti-inflammatory regimens.

Direct trauma to muscle tissue can lead to a contusion, haematoma or myositis ossificans. Myositis generally presents with the classic history of direct trauma and swelling. Heat or ultrasound can exacerbate the process. Imaging studies show calcification within the muscle that should not be in continuity with the bone itself. Biopsies should be performed with caution, as there are case reports in the literature of the active edge of myositis ossificans being misread as osteosarcoma. Even less typically, trauma may result in haemorrhage into an indolent, underlying rhabdomyosarcoma. The clinician should consider alternative diagnoses in patients with hematomata that is slow to resolve or when the history of trauma does not fit with the clinical signs.

The most common of benign soft tissue tumours are ganglion cysts. These cysts are lined by connective tissue, contain mucinous fluid and are found mainly around the wrist, hand, knee and foot. They may be attached to a joint capsule or tendon sheath and may have a connection to the synovial cavity. They are usually asymptomatic but can occasionally cause pain and cosmetic deformity (Chapter 26).² However, ganglion cysts may also represent an important clue regarding other underlying, masquerading disease processes such as osteoarthritis or chronic tendinopathy.

Rheumatological conditions

Inflammatory musculoskeletal disorders in patients presenting to the sports medicine clinic frequently masquerade as a traumatic or mechanical condition. Low back pain of ankylosing spondylitis, psoriatic enthesopathy³ presenting as patellar tendinopathy or flitting arthritis in early rheumatoid arthritis are common examples.

Effective management of athletes presenting with musculoskeletal complaints requires a structured history, physical examination and definitive diagnosis to distinguish soft tissue problems from joint problems and an inflammatory syndrome from a non-inflammatory syndrome. Clues to a systemic inflammatory aetiology may include constitutional symptoms, morning stiffness, elevated acute-phase reactants and progressive symptoms despite modification of physical activity. The mechanism of injury or lack thereof, is also a clue to any underlying disease. In these circumstances, more complete work-up is reasonable, including radiographs, MR imaging and laboratory testing for autoantibodies.⁴

When patients present with an acutely swollen knee without a history of precipitant trauma or patellar tendinopathy without overload, the clinician may be alerted to the possibility that these could be inflammatory in origin. Prominent morning joint or back stiffness, night pain or extra-articular manifestations of rheumatological conditions (e.g. skin rashes, nail abnormalities—Fig. 7.3), bowel disturbance, eye involvement (conjunctivitis, iritis) or urethral discharge may all provide clues.



Figure 7.3 Typical appearance of nails in a patient with psoriatic arthropathy

Inflammation of entheses (e.g. in lateral elbow pain, patellar tendinopathy—Fig. 7.4, insertional Achilles tendinopathy and plantar fasciopathy) is almost universal among those with HLA (human leukocyte antigens) B27-related, seronegative (for rheumatoid factor) arthropathies. Enthesopathy is usually associated with other joint or extra-articular involvement, although a subgroup exists with enthesitis as the sole presentation.²

Muscle disorders

Dermatomyositis and polymyositis are inflammatory connective tissue disorders characterised by proximal limb girdle weakness, often without pain. Dermatomyositis, unlike polymyositis, is also associated with a photosensitive skin rash in light-exposed areas (hands and face). In the older adult, dermatomyositis may be associated with malignancy in approximately 50% of

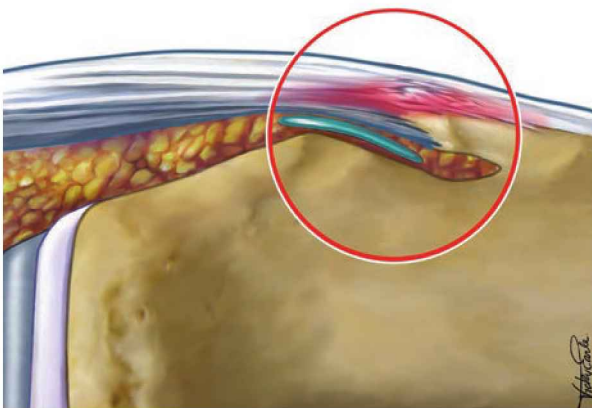


Figure 7.4 Enthesopathy (such as at the distal patellar tendon shown here) should alert the clinician to consider an underlying spondyloarthropathy

cases. The primary malignancy may be easily detectable or occult. In the younger adult, weakness may be profound (e.g. unable to rise from the floor) but in the early stages may manifest only as underperformance in training or competition.

Dermatomyositis and polymyositis may also be associated with other connective tissue disorders such as systemic lupus erythematosus or systemic sclerosis, and muscle abnormality is characterised by elevated creatine kinase levels and electromyographic (EMG) and muscle biopsy changes.

Endurance athletes may complain of myalgia and fatigue that is out of proportion with their training schedule. The differential diagnosis to explain these symptoms is broad. Mitochondrial myopathies, though uncommon, may present with cramping and muscle pain.⁵ Consider myoglobinuria in these patients. Referral to a specialist neurologist for investigations and diagnosis will be necessary.⁶

Regional limb girdle dystrophies such as limb girdle dystrophy and facio-scapulo-humeral dystrophy may also present with proximal limb girdle weakness in young adults. They are also associated with characteristic EMG changes. Ultimately, when an athlete complains of muscle fatigue, aches and pains that fail to respond in a normal time frame expected with the patterns of training, a broader differential diagnosis needs to be considered

Endocrine disorders

Occasionally, the source of the masquerading symptoms in athletes is underlying endocrine disorders and pathologies. Disorders of thyroid function may present with a variety of rheumatological conditions.⁷ Proximal muscle weakness with elevated creatine kinase and fibromyalgia may develop with hypothyroidism. Hyperthyroidism is associated with thyroid acropachy (soft tissue swelling and periosteal bone changes), adhesive capsulitis and painless proximal muscle weakness.⁴ Hyperparathyroidism may be associated with the deposition of calcium pyrophosphate in joints. Patients may develop acute pseudogout or a polyarticular inflammatory arthritis resembling rheumatoid arthritis. X-rays of the wrists or knees may demonstrate chondrocalcinosis of the menisci or triangular fibrocartilage complex (Fig. 7.5).

Adhesive capsulitis or septic arthritis may be the presenting complaint in patients with diabetes mellitus and those with other endocrine disorders such as acromegaly may develop premature osteoarthritis or carpal tunnel syndrome. Patients with hypercalcaemia secondary to malignancy (e.g. of the lung) or other conditions such as hyperparathyroidism can present with bone pain as well as constipation, confusion and renal calculi. A proximal

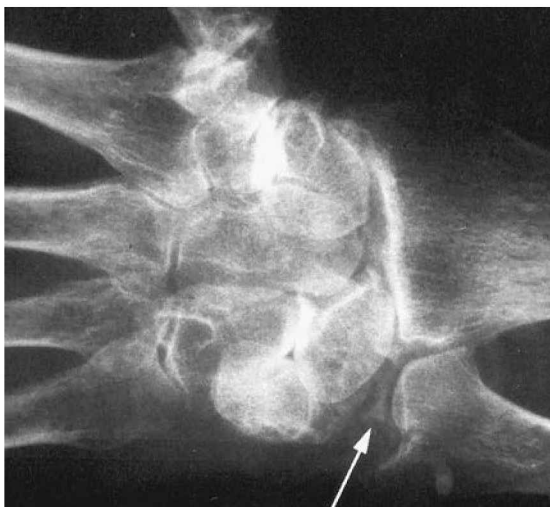


Figure 7.5 Chondrocalcinosis of the triangular fibrocartilage (arrow) in calcium pyrophosphate dehydrate deposition disease

myopathy may develop in patients with primary Cushing's syndrome or after corticosteroid use. When symptoms of myalgia, joint stiffness and swelling do not resolve, a more in-depth laboratory work-up hunting for masqueraders is appropriate and indicated.

Vascular disorders

Patients with venous thrombosis or arterial abnormalities (Fig. 7.6) may present with limb pain and swelling aggravated by exercise. Calf, femoral or axillary veins



Figure 7.6 Angiogram showing common iliac artery stenosis (arrow)

are common sites for thrombosis. While a precipitant cause may be apparent (e.g. recent surgery or air travel), consider also the thrombophilias such as the antiphospholipid syndrome or deficiencies of protein C, protein S, antithrombin III or factor V Leiden. When athletes with no other risk factors for thrombosis develop a blood clot, more extensive work-up is necessary to look for the masquerading underlying cause, which in turn will guide treatment.

The claudicant pain of peripheral vascular disease is most likely to be first noticed with exercise and so patients may present to the sports medicine practitioner. Remember also that arteriopathy can occur in patients with diabetes. Various specific vascular entrapments are also found, such as popliteal artery entrapment, which presents as exercise-related calf pain and thoracic outlet syndrome.

Genetic disorders

Various genetic disorders have been associated with an increased risk of injury and in some cases catastrophic injuries in athletes. Ehlers Danlos with its increased ligamentous laxity has been associated with increased risk of ligament injuries. Osteogenesis imperfect has been associated with an increased fracture risk. Down's syndrome is associated with odontoid hypoplasia with inherent increased risk of cervical spine injuries. Marfan syndrome has been associated with sudden cardiac death. Marfan syndrome is an autosomal dominant disorder of fibrillin characterised by musculoskeletal, cardiac and ocular abnormalities.⁸ Musculoskeletal problems are common due to joint hypermobility, ligament laxity, scoliosis or spondylolysis. In patients with the Marfanoid habitus, referral for echocardiography and ophthalmological opinion should be considered as sudden cardiac death or lens dislocation may result.

Haemochromatosis is an autosomal recessive disorder of iron handling, which results in iron overload. Patients may present with a calcium pyrophosphate arthropathy with characteristic involvement of the second and third metacarpophalangeal joints and hook-shaped osteophytes seen on X-ray of these joints. While ferritin levels are raised in patients with haemochromatosis, it is important to remember that ferritin may also be elevated in athletes taking iron supplements or in response to any acute inflammatory illness.⁹ Fasting transferrin levels and detection of the HFE gene are central to the diagnosis of inherited haemochromatosis. Once again, considering a broad differential diagnosis based on clinical presentation and history may help make these genetic diagnoses, which in turn will allow the clinician to better counsel the athlete regarding risks of participation in certain sports.

Granulomatous diseases

Granulomatous diseases such as tuberculosis and sarcoidosis have been called 'the great mimickers' or great 'masqueraders' secondary to their ability to present in many varied and atypical fashions. They can look like tumours, inflammation, infection or traumatic injuries. Tuberculosis is a granulomatous mycobacterial infection. Musculoskeletal involvement includes chronic septic arthritis and Pott's spine fracture.

Patients with acute sarcoidosis can present with fevers, lower limb (commonly) rash and ankle swelling. The rash of erythema nodosum (Fig. 7.7) may be mistaken for cellulitis and antibiotics have frequently been prescribed in error. The diagnosis is easily made by chest X-ray, which shows changes of bilateral hilar lymphadenopathy. The clinician should remember that the differential diagnosis of bilateral hilar lymphadenopathy includes tuberculosis and lymphoma. Chronic sarcoidosis is a systemic disorder involving the lungs, central nervous system, skin, eyes and musculoskeletal system. Patients can present with chronic arthropathy together with bone cysts or with bone pain due to hypercalcaemia.

Infection

Bone and joint infections, while uncommon, may have disastrous consequences if the diagnosis is missed. Bone pain in children, worse at night or with activity, should alert the clinician to the possibility of osteomyelitis. Bone infection near a joint may result in a reactive joint effusion.

Septic arthritis is rare in the normal joint. In arthritic or diabetic joints or joints that recently underwent arthroscopy, sepsis is much more common. Rapid joint destruction may follow if left untreated.

Even though *Staphylococcus aureus* is the causative organism in more than 50% of cases of acute septic joints, it is imperative that joint aspiration for Gram stain, and

culture and blood cultures are taken before commencement of antibiotic treatment. Once-only or repeated joint lavage may be considered in patients receiving intravenous antibiotic treatment. The immunocompromised patient may present with a chronic septic arthritis. In this situation, tuberculosis or fungal infections should be considered. In suspected cases of septic arthritis, the patient should be admitted to hospital.¹⁰

Another cause of arthritis is Lyme disease, a common arthropod-borne infection in some countries including the US. Hallmarks of Lyme disease are erythema migrans (EM), disruption of electrical conduction of cardiac muscle, the development of neurological abnormalities and episodes of arthritis. Intermittent episodes of arthritis develop several weeks or months after infection and, despite adequate antibiotic therapy, symptoms persist in 10% of patients with arthritis. The severity of arthritis can range from mild to moderate inflammation of the joints and tendons months after infection, to a chronic, debilitating osteoarthropathy complete with destruction of cartilage and erosion of bone in a subset of these individuals within a few years. In severe cases, the highly inflammatory aspects of Lyme arthritis can lead to cartilage and bone erosion with permanent joint dysfunction. The diagnosis of Lyme disease is clinical and serological tests should be used to confirm the clinical diagnosis.¹¹

Pain syndromes

Complex regional pain syndrome type I is a post-traumatic phenomenon characterised by localised pain out of proportion to the injury, vasomotor disturbances, oedema and delayed recovery from injury. The vasomotor disturbances of an extremity manifest as vasodilatation (warmth, redness) or vasoconstriction (coolness, cyanosis, mottling).¹² Early mobilisation, use of motor imagery and avoidance of surgery are important keys to successful management (see also Chapter 6).

Myofascial pain syndromes develop secondary to either acute or overuse trauma. They present as regional pain associated with the presence of one or more active trigger points (Chapter 4).

Fibromyalgia is a chronic pain syndrome characterised by widespread pain, chronic fatigue, decreased pain threshold, sleep disturbance, psychological distress and characteristic tender points. It is often associated with other symptoms including irritable bowel syndrome, dyspareunia, headache, irritable bladder, and subjective joint swelling and pain. Fibromyalgia is diagnosed on the examination finding of 11 of 18 specific tender point sites in a patient with widespread pain. Chronic fatigue syndrome has many similarities to fibromyalgia¹⁰ and may be the same disease process. It may present as excessive post-exercise muscle soreness but is always



Figure 7.7 Erythema nodosum in acute sarcoidosis.
PHOTO COURTESY OF DR RAHEEM B KHERANI

associated with excessive fatigue. There is evidence that exercise improves global wellbeing in women with this condition.¹³

SUMMARY

In summary, it is important for the clinician to be on constant alert for potential masqueraders when dealing with athletic injuries. Clearly, common things are common;

most of the time the clinician will be correct with the first and most likely diagnosis. However, when the history or physical examination is not classic and an athlete fails to improve and heal in the expected time frame of the more common diagnosis, a broader differential diagnosis including masquerading causes should be considered. Stay open minded to the potential if not likelihood of important masquerading pathologies.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e